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Inventor(s)	Ito; Kenichi et al.

Semiconductor device and control method

Abstract

The semiconductor device includes a control unit having redundant processors, a memory storing target data, a secure memory storing a key used for encryption or decryption processing, an cryptographic unit, a secure processor instructing cryptographic processing to the cryptographic unit in response to a request from the control unit, a first bus coupled to the control unit, the memory, the cryptographic unit, and the secure processor, and a second bus coupled to the secure memory, the cryptographic unit, and the secure processor. The control unit communicates with the memory via a predetermined error detection mechanism, the cryptographic unit includes a plurality of cryptographic processors that independently perform cryptographic processing on target data using a key based on an instruction, and each of the plurality of cryptographic processors includes a data transfer unit that performs data transfer with the memory via the error detection mechanism.

Inventors: **Ito; Kenichi (Tokyo, JP), Yamate; Akihiro (Tokyo, JP), Hosotani; Akira (Tokyo, JP)**

Applicant: **RENESAS ELECTRONICS CORPORATION** (Tokyo, JP)

Family ID: **1000008820982**

Assignee: **RENESAS ELECTRONICS CORPORATION (Tokyo, JP)**

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Primary Examiner: Feild; Lynn D

Assistant Examiner: Gavrilenko; Vladimir I

Attorney, Agent or Firm: Rimon P.C.

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This is a Continuation of U.S. patent application Ser. No. 16/573,407 filed on Sep. 17, 2019, which claims the benefit of Japanese Patent Application No. 2018-194009 filed on Oct. 2018 including the specification, drawings and abstract are incorporated herein by reference in their entirety.

BACKGROUND
(1) The present disclosure relates to semiconductor device and methods of controlling the same.
(2) In recent years, in the field of ECUs (Electronic Control Unit), which are examples of semiconductor device, security requirements for preventing threats by malicious third parties in

communication between ECUs have increased in importance. At the same time, communication between ECUs must also meet functional safety requirements to manage the risk of malfunction. As an example of an encryption technique for realizing high-security, Advanced Encryption Standard is known as an example of the encryption technique. Patent Document 1 discloses a technique for realizing a highly reliable storage system by duplicating an AES (Advanced Encryption Standard) circuit and a parity generation unit.
PRIOR-ART DOCUMENT
Patent Document
(3) [Patent Document 1] Japanese Unexamined Publication Laid-Open No. 2009-115857
SUMMARY
(4) However, simply duplicating the AES circuit as described in Patent Document 1 cannot detect a failure of another configuration, and therefore, the functional safety requirements cannot be sufficiently satisfied. Therefore, it is also conceivable to duplicate the entire configuration of the semiconductor device related to the security function, but this increases the size of the security circuit.
(5) Other objects and novel features will become apparent from the description of this specification and the accompanying drawings.
(6) According to one embodiment, in the semiconductor device, the control unit made redundant by the plurality of processors performs communication with the memory via a predetermined error detecting mechanism, the encryption unit includes a plurality of encryption processing units that independently perform encryption processing on target data using a key based on an instruction, and each of the plurality of encryption processing units includes a data transfer unit that performs data transfer with the memory via an error detecting mechanism.
(7) According to the above-mentioned embodiment, it is possible to satisfy both the security requirement and the functional safety requirement while suppressing an increase in the size of circuits in the semiconductor device.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a block diagram showing the concepts of communication between the two semiconductor devices according to a first embodiment.
(2) FIG. **2** is a diagram showing a configuration of a semiconductor device according to the first embodiment;
(3) FIG. **3** is a block diagram showing an internal configuration of an encryption unit according to the first embodiment;
(4) FIG. **4** illustrates an example of data stored in a transmitter memory according to the first embodiment.
(5) FIG. **5** is a sequence diagram illustrating a flow of encryption, verification and transmission processing of data according to the first embodiment.
(6) FIG. **6** is a sequence diagram illustrating a flow of encryption, verification and transmission processing of data according to the first embodiment.
(7) FIG. **7** illustrates an example of data stored in a receiver memory according to the first embodiment.
(8) FIG. **8** is a sequence diagram showing a flow of receiving, decrypting and verifying data according to the first embodiment.
(9) FIG. **9** is a sequence diagram showing a flow of receiving, decrypting and verifying data according to the first embodiment.
(10) FIG. **10** is a diagram showing a configuration of a semiconductor device according to a second

embodiment;
(11) FIG. **11** is a block diagram showing an internal configuration of an encryption unit according to the second embodiment;
(12) FIG. **12** illustrates an example of data stored in a receiver memory according to the second embodiment.
(13) FIG. **13** is a sequence diagram showing a flow of receiving, decrypting and verifying data according to the second embodiment.
(14) FIG. **14** is a sequence diagram showing a flow of receiving, decrypting and verifying data according to the second embodiment.

DETAILED DESCRIPTION

(15) For clarity of explanation, the following description and drawings are appropriately omitted and simplified. In addition, the elements described in the drawings as functional blocks for performing various processes can be configured as CPUs (Central Processing Unit), memories, and other circuits in terms of hardware, and are realized by programs loaded into the memories in terms of software. Therefore, it is understood by those skilled in the art that these functional blocks can be realized in various forms by hardware alone, software alone, or a combination thereof, and the present invention is not limited to any of them. In the drawings, the same elements are denoted by the same reference numerals, and a repetitive description thereof is omitted as necessary.
(16) Also, the programs described above may be stored and provided to a computer using various types of non-transitory computer readable media. Non-transitory computer readable media includes various types of tangible storage media. Examples of non-transitory computer-readable media include magnetic recording media (e.g., flexible disks, magnetic tapes, hard disk drives), magneto-optical recording media (e.g., magneto-optical disks), CD-ROM (Read Only Memory, a CD-R, a CD-R/W, solid-state memories (e.g., masked ROM, PROM (Programmable ROM), EPROM (Erasable PROM, flash ROM, RAM (Random Access Memory)). The program may also be supplied to the computer by various types of transitory computer-readable media. Examples of transitory computer-readable media include electrical signals, optical signals, and electromagnetic waves. The transitory computer readable medium may provide the program to the computer via wired or wireless communication paths, such as electrical wires and optical fibers.
(17) Here, the following art according to the present embodiment will be supplemented. One approach to confirming the authenticity of messages in communication is to assign a message authentication code (Message Authentication Code: MAC) to the sender of the data (message) and confirm (verify) the MAC to the receiver. In addition, the authenticity of messages has been mainstream as security requirements in the communication of ECUs (Electronic Control Unit) for automobiles. In recent years, the security requirement has been required not only for authenticity but also for confidentiality. Therefore, as a technique for simultaneously satisfying the authenticity and confidentiality of messages, there are cryptographic algorithms with authenticity such as AES-GCM (Galois/Counter Mode). The AES-GCM is a technique for performing encryption of a message and authenticating the message at the same time by using a symmetric key.
(18) In addition, in communication between ECUs mounted in automobiles, it is required to check whether communication data has failed or not in an end-to-end (End to End) manner in order to satisfy functional safety requirements. Examples of safety mechanisms for end-to-end detection of communication data for faults include CRC (Cyclic Redundancy Check) and ECC (Error Correction Code)/EDC (Error Detection Code). However, a method of providing and confirming the MAC described above can also be used for failure detection. Therefore, when a method of assigning and confirming MACs is applied to secure IP (Intellectual Property) in a semiconductor device having high security requirements, the secure IP itself can be treated as one of security mechanisms.
(19) Here, some problems to be solved by the present embodiment will be described. First, when data for satisfying functional safety requirements is communicated between ECUs, if an

authenticated encryption such as AES-GCM is applied to the data by secure IP, failure detection may not be possible end-to-end. The reason for this is that, in the encryption with authentication, since encryption and tag generation or decryption and tag verification are performed simultaneously, there is a possibility that communication is established when a failure occurs. For example, even if encryption is performed incorrectly due to a failure of the encryption engine, a tag corresponding to encrypted data is generated due to an incorrect encryption process. In this case, on the receiving side, the tag correctly generated from the erroneously encrypted data coincides with the tag generated on the transmitting side corresponding to the erroneously encrypted data added to the received data. Therefore, on the receiving side, the tag verification succeeds, and the decoding process can be performed normally, so that there is a case where the communication is established due to erroneous data.
(20) Alternatively, even if the decryption is an incorrect treatment (incorrect decryption) due to the failure of the cryptographic engine, if the tag generated for verification from the received data is correct, the verification of the tag assigned to the received data is succeeded. Therefore, even if the decoded data is erroneous, communication may be established.
(21) Further, in the case of the method of detecting and correcting errors by duplicating the AES circuit and parity generation as described in Patent Document 1, only the cryptographic engine is duplicated. However, in this case, since the CPU (secure CPU) in the secure IP performs data transfer between the memory and the cryptographic engine outside the secure IP, the failure of the data communication intervened by the secure CPU or the failure of the secure CPU itself cannot be detected.
(22) Here, as a security mechanism, it is considered that the entire secure IP including the secure CPU and the cryptographic engine is duplicated. However, in this case, the influence on the cost due to the increase in the chip size becomes large. In other words, if the security requirements and the functional safety requirements are satisfied in the semiconductor device, there arises a problem that the circuit size is increased.
(23) In addition, a countermeasure for performing failure detection by attaching a CRC to the target data before the target data is processed by the authenticated encryption may be considered. However, in this case, since the calculation processing of CRC is increased as compared with the case where only the authenticated encryption is applied, the overhead of the transmission/reception processing is increased. In addition, it is also conceivable that the processor performs the same operation such as the cryptographic processing twice, which is duplication by software. However, even in this case, the overhead of the transmission/reception processing becomes large.
(24) From the above, an embodiment for solving at least one of these problems will be described below.
First Embodiment
(25) FIG. **1** is a diagram showing concepts of communication between two semiconductor devices **1** and **2** according to the first embodiment. The semiconductor devices **1** and **2** are connected to each other by a signal line **3**, and data can be communicated with each other. It is assumed that the semiconductor devices **1** and **2** can encrypt (encryption) and decrypt (decryption) the data to be communicated by using a common key held in advance by both of them by using predetermined cryptographic process (cryptographic processing) algorithms. It is also assumed that the cryptographic processing algorithm can generate and verify a MAC from the target data. It is also assumed that the semiconductor devices **1** and **2** have the same function and can interchange the transmitting side and the receiving side with each other. The semiconductor devices **1** and **2** are, for example, microcomputers, MCUs (Micro Control Unit), SoCs (System on Chip), ECUs, and the like, and may be mounted on the same or different automobiles. The semiconductor devices **1** and **2** may also be referred to as semiconductor chips.
(26) Here, it is assumed that the semiconductor device **1** encrypts the data to be communicated by a predetermined encryption algorithm using the common key to generate encrypted data **421**, and

also generates an authentication tag **422**. The authentication tag **422** is an example of a message authentication code. Then, the semiconductor device **1** transmits the communication data **42** including the encrypted data **421** and the authentication tag **422** to the semiconductor device **2**, which is the receiving side, via the signaling line **3**. Then, the semiconductor device **2** receives the communication data **42** via the signal line **3**, decrypts the encrypted data **421** included in the communication data **42** by a predetermined cryptographic process algorithm using the common key to generate decrypted data (not shown), and also generates verification tag (not shown). Then, the semiconductor device **2** compares the verification tag generated by itself with the authentication tag **422** included in the received communication data **42**, and when the verification is performed, it determines that the decrypted data has been normally received, and uses the decrypted data for subsequent processing.
(27) FIG. **2** is a diagram showing the configuration of the semiconductor device **1** according to the first embodiment. Semiconductor device **1** supports security requirements and functional safety requirements. The semiconductor device **1** includes a secure CPU **111**, a secure memory **112**, a cryptographic unit **113**, a control unit **121**, a memory **122**, and a communication interface (IF) **123**. Here, the secure CPU **111**, the secure memory **112**, and the cryptographic unit **113** belong to the secure area **11**, and the control unit **121**, the memory **122**, and the communication IF **123** belong to the non-secure area **12**. The secure CPU **111** belonging to the secure area **11**, the secure memory **112**, and the cryptographic unit **113** may be collectively referred to as secure IP. The secure CPU **111**, the cryptographic unit **113**, the control unit **121**, the memory **122**, and the communication IF **123** except for the secure memory **112** are communicably connected via the main bus **13**. The secure CPU **111**, the secure memory **112**, and the cryptographic unit **113** are communicably connected via a second bus **14**. That is, the control unit **121** or the like of the non-secure area **12** cannot directly access the secure memory **112** of the secure area **11**.
(28) The secure CPU **111** is a processor that instructs the cryptographic unit **113** to perform an encryption process in response to a request from the control unit **121**. The secure CPU **111** reads the key from the secure memory **112** and sets the key in the cryptographic unit **113** at the time of instructing the encryption process. The secure CPU **111** sets the setting information including the input/output addresses of the memories **122** and the types of cryptographic processing algorithms included in the requests in the cryptographic unit **113** at the time of instructing the cryptographic processing. The secure CPU **111** stores the setting information in the memory **122**.
(29) The secure memory **112** is a volatile or nonvolatile storage device. Secure memory **112** includes an ECC/EDC **1121** and storage area **1122**. The ECC/EDC **1121** is an example of an error detecting mechanism. Here, the “error detection mechanism” is a system including “error detection” or “error detection and error correction”. Therefore, the ECC/EDC **1121** may detect or correct an error with respect to the data in the storage area **1122**.
(30) The storage area **1122** stores the key **1123**. The key **1123** is key data used for encryption processing for encryption or decryption. Here, it is assumed that the key **1123** is a common key between semiconductor devices **1** and **2**.
(31) The cryptographic unit **113** performs encryption processing. Here, the cryptographic process is, for example, a process using an authenticated cryptographic algorithm such as AES-GCM, and includes data encryption and tag generation, or data decryption and tag verification. However, the authenticated cryptographic algorithms are not limited to AES-GCM. The internal configuration of the cryptographic unit **113** will be described later.
(32) The control unit **121** is a functional block made redundant by a plurality of processors. It is assumed that the control unit **121** is made redundant by CPU **1211** and CPU **1212** in accordance with the DCLS (Dual Core Lock Step) system. Therefore, the control unit **121** satisfies the functional safety requirements. The CPU **1211** has the ECC/EDC **12110**, and can detect or correct an error in transmission/reception data to/from the ECC/EDC of another configuration via the main bus **13** using the ECC/EDC **12110**. In addition, the CPU **1212** has the ECC/EDC **12120**, and can

similarly detect or correct errors in transmission/reception data. One of the CPU **1211** and **121** is a master and the other is a checker. The checker executes the same processing as the master several clocks after the processing of the master. The master accesses to other components via the main bus **13** when both processing results by the master and checker are matched. However, the control unit **121** may have a redundancy configuration by another method.

(33) The memory **122** is a volatile or nonvolatile storage device. The memory **122** includes an ECC/EDC1220 and a storage area (not shown). Therefore, it can be said that the control unit **121** communicates with the memory **122** via the ECC/EDC.

(34) The communication IF **123** is an interface that communicates with the semiconductor device **2** outside the semiconductor device **1**, for example, via the signal line **3**.

(35) It should be noted that CRCs may be employed instead of ECC/EDC as error detecting mechanisms used in the above-described configurations.

(36) FIG. **3** is a block diagram showing an internal configuration of the cryptographic unit **113** according to the first embodiment. The cryptographic unit **113** includes an cryptographic processors **31** and **32**, and a comparison unit **33**. Each of the cryptographic processors **31** and **32** independently performs cryptographic processing on the target data stored in the memory **122** by using the key **1123** based on the cryptographic processing instruction from the secure CPU **111**.

(37) The cryptographic processor **31** is a master cryptographic engine, and includes a storage unit **311**, an ECC/EDC **312**, a data-transfer unit **313**, and a cryptographic circuit **314**. The storage unit **311** is a storage area such as a register, and stores setting information **3111** and a key **3112** set by the secure CPU **111**. The setting information **3111** includes an address which is an access destination in the memory **122**, and a type of an arithmetic algorithm of the cryptographic process. The types may be, for example, an AES-GCM calculation mode or types of calculation algorithms. The key **3112** is set with the above-described key **1123**, and is a common key. It is assumed that the storage unit **311** has a mechanism for copying data to be stored or updated to the storage unit **321** in the cryptographic processor **32**. Instead of this mechanism, however, the secure CPU **111** may store the same data in both of the storage units **311** and **321**.

(38) The ECC/EDC **312** can detect or correct errors in transmission/reception data to/from the ECC/EDC of another component via the main bus **13**. The data transfer unit **313** performs data transfer with another component directly via the main bus **13** without using the secure CPU **111**.

(39) The cryptographic circuit **314** uses the key **3112** to perform the AES-GCM operation in the operation mode corresponding to the type included in the setting information **3111**. The cryptographic circuit **314** performs the AES-GCM operation (cryptographic process) on the target data acquired from the data transfer circuit **313**. At this time, in the case of encryption, the cryptographic circuit **314** encrypts the target data and generates the authentication tag as encryption processing. In the case of decryption, the cryptographic circuit **314** decrypts the target data and generates a verification tag as cryptographic processing. Then, the cryptographic circuit **314** outputs the processing result of the cryptographic processing to the comparison unit **33**. Here, the processing result is encrypted data and the authentication tag in the case of encryption, and is decrypted data and the verification tag in the case of decryption. In addition, the cryptographic circuit **314** outputs the result to the data transfer unit **313** and stores the result in the memory **122** via the ECC/EDC **312**.

(40) The cryptographic processor **32** is a checker cryptographic engine, and includes a storage unit **321**, an ECC/EDC **322**, a data-transfer unit **323**, and a cryptographic circuit **324**. Since the storage unit **321**, the ECC/EDC **322**, the data-transfer unit **323**, and the cryptographic circuit **324** are generally the same as those in the cryptographic processor **31**, the difference from the cryptographic processor **31** will be described here. The storage unit **321** stores setting information **3211** and a key **3212**. Here, as described above, the setting information **3211** and the key **3212** are data stored or updated in the storage unit **311**, that is, the same data as the setting information **3111** and the key **3112**.

(41) The cryptographic circuit **324** notifies the comparison unit **33** of its own processing result, but unlike the cryptographic circuit **314** because it is for the checker, it does not output its own processing result to the main bus **13** through the data transfer unit **313**.
(42) The comparison unit **33** is an example of a failure detection circuit, and detects a failure of either of the cryptographic processors **31** and **32** by comparing the processing results of the cryptographic processing of the cryptographic circuits **314** and **324**. The comparison unit **33** determines that there is no failure (failure is not detected) when the processing results match, and that there is a failure (failure is detected) when the processing results do not match. When detecting a failure, the comparison unit **33** outputs the fact to at least the outside of the cryptographic unit **113**. For example, the comparison unit **33** outputs the fact that the failure has been detected to the secure CPU **111**, the control unit **121**, or the outside of the semiconductor device **1**. When detecting a failure, the comparison unit **33** may notify the cryptographic circuits **314** and **324**.
(43) FIG. **4** is a diagram showing examples of data stored in the memory **122** in the transmission-side semiconductor device **1** according to the first embodiment. The memory **122** stores a functional safety related data **41**, the encrypted data **421**, the authentication tag **422**, the request information **43**, and the setting information **44**. The functional safety related data **41** includes plaintext **411**, IV (Initialization Vector) **412**, and AAD (Additional Authenticated Data) **413**, which are data to be transmitted. Note that IV **412** and AAD **413** are examples of data used for AES-GCM, and are not limited to these data. The encrypted data **421** and the authentication tag **422** are processing results of encryption processing (encryption and tag generation) by the cryptographic circuit **314**. The request information **43** is information used when the control unit **121** requests the secure CPU **111** to encrypt the plaintext **411**. The request information **43** includes, for example, an address in the memory **122** in which the functional safety related data **41** is stored, and an address in the memory **122** for storing the encrypted data **421**, the authentication tag **422**, and the setting information **44**. That is, the request information **43** includes input/output addresses for the data transfer units **313** and **323** to access the memory **122**. The request information **43** includes the type of the calculation algorithm of the cryptographic process. The setting information **44** stores the setting information **3111** or **3211** set in the cryptographic unit **113** by the secure CPU **111** in the memory **122**.
(44) FIGS. **5** and **6** are sequence diagrams showing a flow of data encryption, verification, and transmission processing according to the first embodiment. First, the control unit **121** stores the functional safety related data **41** including the plaintext **411**, which is data to be transmitted, in the S**101** unit **122**. Next, the control unit **121** makes an encryption request for requesting the secure CPU**111** to encrypt the plaintext **411** and generate the authentication tag (S**102**). The encryption request includes request information **43**. The control unit **121** may store the request information **43** in the memory **122** at this timing.
(45) The secure CPU **111** sets a key to the cryptographic unit **113** (S**103**). Specifically, the secure CPU **111** reads the key **1123** from the secure memory **112** in response to the encryption request from the control unit **121**, and sets, that is, stores, the key **1123** in the cryptographic unit **113**. At this time, the secure CPU **111** stores the key **1123** at least in the storage unit **311** of the cryptographic processor **31**. Then, for example, the key **3112** stored in the storage unit **311** is copied to the storage unit **321** as the key **3212** by the above-described mechanism.
(46) Subsequently, the secure CPU **111** issues an encryption instruction to the cryptographic unit **113** in response to the encryption request from the control unit **121** (S**104**). Specifically, the secure CPU **111** extracts the input/output addresses of the memories **122** and the types of calculation algorithms of the encryption process from the request information included in the encryption request, and puts the extracted addresses into the encryption instruction as setting information. The secure CPU **111** stores the setting data **3111** at least in the storage unit **311** of the cryptographic unit **31**. Then, for example, the setting information **3111** stored in the storage unit **311** is copied to the storage unit **321** as the setting information **3211** by the above-described mechanism.

(47) Thereafter, the cryptographic processors **31** and **32** perform encryption processing independently of each other in response to the encryption instruction from the secure CPU **111**. Although the cryptographic processors **31** and **32** are described in this order, they may be processed in parallel or may be different by several clocks.
(48) First, the cryptographic processor **31** reads out the functional safety related data **41** from the memory **122** in response to the encryption instruction (S**105**). Specifically, the cryptographic processor **31** accesses the input address contained in the setting information **3111** via the main bus **13** using the data transfer unit **323** and the ECC/EDC **312**, reads out the function safety-related data **41** from the memory **122**, and stores the data in a register, etc. in the cryptographic processor **31**.
(49) Next, the cryptographic processor **31** performs encryption and generation of authentication tags (S**106**). Specifically, the cryptographic circuit **314** reads the type of the cryptographic operation algorithm included in the setting information **3111** from the storage unit **311**, and specifies the AES-GCM operation mode. The cryptographic circuit **314** reads the key **3112** from the storage unit **311**, and extracts the IV **412** and the AAD **413** from the functional safety-related data **41** acquired by the data transfer unit **313**. Then, the cryptographic operation circuit **314** uses the key **3112**, the IV**412**, and the AAD**413** to calculate the AES-GCM according to the specified operation mode for the plaintext **411** included in the functional safety-related data **41**. At this time, the cryptographic circuit **314** encrypts the plaintext **411** to generate encrypted data, and also generates an authentication tag. Then, the cryptographic circuit **314** outputs the generated encrypted data and the authentication tag to the comparison unit **33** as a processing result.
(50) Similarly, in response to the encryption instruction, the cryptographic processor **32** reads out the function safety-related data **41** from the memory **122** (S**107**), performs encryption and generation of the authentication tag (S**108**), and outputs the processing result to the comparison unit **33**.
(51) Thereafter, the comparison unit **33** compares the processing results inputted from the cryptographic processors **31** and **32**, respectively, and determines whether they coincide with each other or not (S**109**). When it is determined that the processing results do not coincide with each other, the comparison unit **33** outputs a message indicating that a failure has occurred to the outside of the cryptographic unit **113**. The comparison unit **33** may also output the fact that a failure has occurred to the cryptographic circuit **314**. Further, the comparison unit **33** may output a message indicating the presence of a failure to the cryptographic circuit **324**.
(52) The cryptographic circuit **314** outputs the encrypted data and the authentication tag to the data transfer unit **313** and stores them in the memory **122** via the ECC/EDC **312** (S**111**). After the encryption process by the cryptographic unit **113**, the secure CPU **111** reads the setting information from the cryptographic unit **113** and stores the setting information in the memory **122** (S**112**). For example, the secure CPU **111** reads the setting information **3111** from the storage unit **311** of the cryptographic processor **31**, and stores the setting information **3111** as the setting information **44** at the address outputted from the memory **122** via the main bus **13**.
(53) Thereafter, the control unit **121** reads out the request information **43** and the setting information **44** from the memory **122** (S**113**), compares the request information **43** and the setting information **44** (S**114**), and determines whether they coincide with each other or not. When it is determined that they coincide with each other, the control unit **121** determines that the secure CPU **111** has performed the encryption process as requested by the request data **43**. If they do not coincide in the S**114** of steps, a failure of the secure CPU **111** can be detected. That is, the functional safety requirements can be met without redundancy of the secure CPU **111** and without increasing the circuitry size for the secure CPU **111**.
(54) When it is determined that the secure CPU **111** has directed encryption processing as requested, the control unit **121** sends the encrypted data and the authentication tag to the semiconductor device **2** (S**115**). Specifically, the control unit **121** reads the encryption data **421** and the authentication tag **422** from the memory **122** as the communication data **42**, and outputs the

communication data **42** to the communication IF **123** with the semiconductor device **2** as the destination. As a result, the communication IF **123** transmits the communication data **42** to the semiconductor device **2** via the signal line **3**.
(55) Since the ECC/EDC is used for data transfer in steps S**105**, the S**107**, and the S**111**, if a failure occurs in the data transfer unit **313** or **324**, it can be detected.
(56) Next, the semiconductor device **2** which receives the communication data **42** will be described. First, since the semiconductor device **2** has the same configuration as the semiconductor device **1**, the illustration thereof is omitted, the same reference numerals are given in the following explanation, and the following explanation will focus on the differences from the semiconductor device **1**.
(57) FIG. **7** is a diagram showing examples of data stored in the memory **122** in the receiving-side semiconductor device **2** according to the first embodiment. The memory **122** stores the received data the processing result **46**, the request information **47**, and the setting information **48**. The received data **45** is communication data **42** received by the semiconductor device **2** from the semiconductor device **1** via the signaling line **3**. The received data **45** includes an encrypted data **451** and an authentication tag **452**. The encrypted data **451** and the authentication tag **452** are the same data as the encrypted data **421** and the authentication tag **422** described above. The processing result **46** includes decoded data **461** and a verification tag **462**. The processing result **46** is a processing result of cryptographic processing (decryption and generation of verification tag) by the cryptographic circuit **314** in the cryptographic unit **113** of the semiconductor device **2**. The request information **47** is information used when the control unit **121** of the semiconductor device **2** requests the secure CPU **111** to decrypt the encrypted data **451** (decryption and generation of verification tags). The request information **47** may include the same information as the request information **43** described above. The setting information **48** stores the setting information **3111** or **3211** set in the cryptographic unit **113** by the secure CPU **111** of the semiconductor device **2** in the memory **122**.
(58) FIGS. **8** and **9** are sequence diagrams showing a flow of data reception, decoding, and verification processing according to the first embodiment. First, the communication IF **123** receives the communication data **42**, i.e., the encrypted data **421** and the authentication tag **422**, from the semiconductor device **1** via the signal line **3**, and stores the received data **45** in the memory **122** via the main bus **13** (S**201**). The control unit **121** may receive the received data **45** from the communication IF **123** and store it in the memory **122**.
(59) Next, the control unit **121** makes a decryption request for requesting the secure CPU **111** to decrypt the encrypted data **451** and generate verification tag (S**202**). The decryption request includes request information **47**. The control unit **121** may store the request information **47** in the memory **122** at this timing.
(60) The secure CPU **111** sets a key to the cryptographic unit **113** (S**203**) in the same manner as the above-described S**103** of steps.
(61) Next, the secure CPU **111** issues a decryption instruction to the cryptographic unit **113** in response to a decryption request from the control unit **121** (S**204**). More specifically, the secure CPU **111** extracts the input/output addresses of the memory **122** and the types of calculation algorithms of the cryptographic process from the request information included in the decryption request, and includes them as setting information in the decryption instruction. The setting of the setting information is the same as the treatment in the cryptographic unit **113** according to the encryption instruction.
(62) Thereafter, the cryptographic processors **31** and **32** perform cryptographic processing independently of each other in response to a decryption instruction from the secure CPU **111**. Although the cryptographic processors **31** and **32** are described in this order, they may be processed in parallel or may be different by several clocks.
(63) First, the cryptographic processor **31** reads the encrypted data **451** from the memory **122** in

response to the decryption instruction (S**205**). Next, the cryptographic processor **31** performs decryption and generation of verification tag (S**206**). Specifically, the cryptographic circuit **314** reads the type of the cryptographic operation algorithm included in the setting information **3111** from the storage unit **311**, and specifies the AES-GCM operation mode. The cryptographic circuit **314** reads the key **3112** from the storage unit **311**. Then, the cryptographic circuit **314** performs AES-GCM operation on the encrypted data **451** in accordance with the specified operation modes using the key **3112**. At this time, the cryptographic circuit **314** decrypts the encrypted data **451** to generate decrypted data, and also generates a verification tag. Then, the cryptographic circuit **314** outputs the generated decrypted data and the verification tag to the comparison unit **33** as a processing result.
(64) Similarly, in response to the decryption instruction, the cryptographic processor **32** reads the encrypted data **451** from the memory **122** (S**207**), performs decryption and generation of verification tag (S**208**), and outputs the processing result to the comparison unit **33**.
(65) Thereafter, the comparison unit **33** compares the processing results inputted from the cryptographic processors **31** and **32**, respectively, and determines whether or not they coincide with each other (S**209**). When it is determined that the processing results do not coincide with each other, the comparison unit **33** outputs a message indicating that a failure has occurred to the outside of the cryptographic unit **113**. The comparison unit **33** may also output the fact that a failure has occurred to the cryptographic circuit **314**. Further, the comparison unit **33** may output the presence or absence of a failure to the cryptographic circuit **324**.
(66) The cryptographic circuit **314** outputs the decryption data and verification tag generated by the cryptographic circuit to the data transfer unit **313** and stores them in the memory **122** via the ECC/EDC **312** (S**211**). After the decryption process by the cryptographic unit **113**, the secure CPU **111** reads the setting information from the cryptographic unit **113** and stores the setting information in the memory **122** (S**212**).
(67) Thereafter, the control unit **121** reads out the request information **47** and the setting information **48** from the memory **122** (S**213**), compares the request information **47** and the setting information **48** (S**214**), and determines whether or not they coincide with each other. When it is determined that they coincide with each other, the control unit **121** determines that the secure CPU **111** instructs the cryptographic process as requested by the request data **47**. If they do not coincide in the S**214** of steps, a failure of the secure CPU **111** can be detected.
(68) When it is determined in the step S**214** that the request information **47** and the setting information **48** coincide with each other, the control unit **121** reads the authentication tag **452** and the verification tag **462** from the memory **122** (S**215**), compares the authentication tag **452** and the verification tag **462** with each other (S**216**), and determines whether or not the authentication tag **452** and the verification tag **462** coincide with each other. If it is determined that they match, the control unit **121** determines that the received data **45** has been authenticated (the authentication result has passed). Therefore, the control unit **121** can proceed with the subsequent processing using the decoded data **461**. On the other hand, if they do not coincide with each other in the S**216** of steps, the received data cannot be authenticated, so that the control unit **121** discards the decrypted data **461** and the like.
(69) As described above, in the present embodiment, the secure CPU **111** belonging to the secure area **11** is excluded from the DCLS, but the security requirements and the functional safety requirements can be satisfied. Therefore, as compared with the case where the entire secure region **11** is duplicated, an increase in the circuit scale can be suppressed, and functional safety can be coped with at low cost.
(70) Further, the control unit **121** does not apply or confirm the error correction code by the CRC to the data to be encrypted. Therefore, it is possible to suppress the overhead of processing the CRC. In addition, there is no need for duplication or verification of cryptographic processing (encryption or decryption) by software. Thus, the overhead of software processing can be suppressed.

Therefore, since the transmitting side and the receiving side can complete the cryptographic processing in one processing, respectively, the data processing can be performed with high efficiency.
(71) As described above, in the present embodiment, the cryptographic engine part (cryptographic unit **113**) is DCLS encrypted, and the respective cryptographic engines are provided with data-transferring units, and ECC/EDC is given to the data-transferring units. Therefore, the application of the security service to the functional safety-related data can be implemented at low cost and with high efficiency.
(72) The present embodiment can also be understood as follows. First, the present embodiment can detect a failure in the secure area **11** in the semiconductor device. As a means for this purpose, the cryptographic processor is made redundant, and the processing results of the respective cryptographic processing units are compared by the comparison unit. This makes it possible to detect a failure in the cryptographic processing by the cryptographic engine.
(73) As another method, data transfer units are provided in the cryptographic engines, and direct data transfer is performed with the memory **122** in the non-secure area **12** by using the ECC/EDC (without using the secure CPUs). As a result, it is possible to detect a communication failure between the secure area **11** side and the memory **122** on the non-secure area **12** side.
(74) As another means, request information when the processor on the non-secure area **12** side requests encryption processing to the secure CPU on the secure area **11** side is held in the memory **122**. Then, the setting information when the secure CPU instructs the encryption engine is stored in the memory **122** on the non-secure area **12** side. Thereafter, the processor on the non-secure area **12** side verifies the secure CPU by comparing the setting information with the request information. As a result, it is possible to detect a failure without making the secure CPU redundant.
(75) The present embodiment can also be expressed as follows. That is, the semiconductor device according to the present embodiment includes a control unit, a memory, a secure memory, an encryption unit, a secure processor, a first bus, and a second bus. Here, the control unit is made redundant by a plurality of processors. The memory stores target data. The secure memory stores a key used for cryptographic processing for encryption or decryption. The cryptographic unit performs the cryptographic processing. The secure processor instructs the encryption unit to perform the encryption process in response to a request from the control unit. A first bus communicatively connects the controller, the memory, the cryptographic unit, and the secure processor. A second bus communicatively connects the secure memory, the cryptographic unit, and the secure processor. The controller communicates with the memory via a predetermined error detection mechanism. The cryptographic unit includes a plurality of cryptographic processors that independently perform the cryptographic processing on the target data using the key based on the instruction. Each of the plurality of cryptographic processors includes a data transfer unit that performs data transfer with the memory via the error detection mechanism.
(76) As a result, it is possible to detect the fraud of the communication between the secure area and the non-secure area without redundancy of the secure processor. Since the cryptographic processors is made redundant, for example, even if the result of the cryptographic processing is compared and verified by the control unit, it is possible to detect a failure of the cryptographic processing. In addition, an increase in the circuit scale can be suppressed as compared with redundancy of the entire secure area.
(77) The cryptographic unit may further include a failure detection circuit for comparing the treatment results of the cryptographic processing of each of the plurality of cryptographic processing units to detect a failure of any of the cryptographic processors. As described above, by performing comparison by hardware processing, it is possible to speed up detection of faults between redundant cryptographic engines.
(78) Further, the first cryptographic processor of the plurality of cryptographic processors may store the processing result of its own cryptographic processing in the memory using the data

transfer unit. The other cryptographic processor among the plurality of cryptographic processors may discard the processing result of their own cryptographic processing. As a result, only the processing result of the encryption engine of the master is transferred, and the transfer amount can be minimized.
(79) Further, it is preferable that the secure processor stores setting information relating to the cryptographic processing set in the cryptographic unit in the instruction in the memory, and the control unit detects a failure of the secure processor based on request information used for the request to the secure processor and the setting information stored in the memory. Thus, a failure can be detected without redundancy of the secure processor.
(80) Further, the secure processor may include an address, which is an access destination in the memory included in the request information, in the setting information and may set the address in the encryption unit, and the data transfer unit may access the address in the memory. As a result, data transfer can be efficiently performed.
(81) When the target data is data to be transmitted to another semiconductor device, it is preferable that each of the plurality of cryptographic processors performs encryption and generation of a message-authenticating code for the target data using the key as the cryptographic processing based on the instruction. This makes it possible to satisfy security requirements and functional security requirements for cryptographic processing at the time of transmission.
(82) When the target data is received data from another semiconductor device, it is preferable that each of the plurality of cryptographic processors decrypt the target data using the key and generate a first message-authenticating code as the cryptographic processing based on the instruction. This makes it possible to satisfy security requirements and functional security requirements for cryptographic processing at the time of reception.
(83) It is further assumed that the received data includes encrypted data encrypted from predetermined data using the key in the other semiconductor device and a generated second message-authenticating code. In this case, at least the first cryptographic processor of the plurality of cryptographic processor may store the decrypted data and the first message authentication code in the memory using the data transfer unit. The controller may compare the first message authentication code and the second message authentication code stored in the memory to detect a failure of the first cryptographic processor. As a result, message authentication can be performed more appropriately.
(84) The present embodiment can also be expressed as follows. That is, the semiconductor device according to the present embodiment includes a control unit, a memory, a secure memory, an cryptographic unit, a secure processor, a first bus, and a second bus. Here, the control unit is made redundant by a plurality of processors. The memory stores target data. The secure memory stores a key used for cryptographic processing for encryption or decryption. The cryptographic unit performs the cryptographic processing. The secure processor instructs the encryption unit to perform the encryption process in response to a request from the control unit. A first bus communicatively connects the controller, the memory, the cryptographic unit, and the secure processor. A second bus communicatively connects the secure memory, the cryptographic unit, and the secure processor. The controller communicates with the memory via a predetermined error detection mechanism. The cryptographic unit includes a plurality of cryptographic processors that independently perform the cryptographic processing on the target data using the key based on the instruction. The secure processor stores, in the memory, setting information relating to the cryptographic processing set in the cryptographic unit in the instruction. The control unit detects a failure of the secure processor based on the request information used for the request to the secure processor and the setting information stored in the memory.
(85) As a result, it is possible to detect a failure in communication between the secure area and the non-secure area without redundancy of the secure processor. Since the cryptographic processor is made redundant, for example, even if the result of the cryptographic processing is compared and

verified by the control unit, it is possible to detect a failure of the cryptographic processing. In addition, an increase in the circuit scale can be suppressed as compared with redundancy of the entire secure area.

(86) Alternatively, the present embodiment can be expressed as follows. That is, the semiconductor device control method according to the present embodiment is a method for controlling the above-described semiconductor device. The secure processor instructs the cryptographic unit to perform the cryptographic process in response to a request from the control unit. Each of the plurality of cryptographic processors acquires the target data from the memory via the error detection mechanism based on the instruction, and performs the cryptographic processing on the target data using the key. Thereafter, at least one of the plurality of cryptographic processors stores the processing result of its own cryptographic processing in the memory via the error detection mechanism. This achieves the same effect as described above.

Second Embodiment

(87) The second embodiment is an improvement example of the first embodiment described above. In the second embodiment, a tag comparator is added to the cryptographic processor. The tag comparator itself does not necessarily need to be redundant. That is, at least the first cryptographic processor of the plurality of cryptographic processors further includes a comparator for comparing the second message authentication code acquired from the memory using the data transfer unit with the first message authentication code. It is assumed that the received data includes encrypted data encrypted from predetermined data using the key in the other semiconductor device and the generated second message-authenticating code. As a result, the comparison processing between the authentication tag **452** and the verification tag **462** on the non-secure area **12** side becomes unnecessary, and the processing load of the software by the CPU can be reduced as compared with the first embodiment.

(88) Further, the first cryptographic processor may store the decrypted data and the comparison result by the comparator in the memory using the data transfer unit, and the control unit may discard the decrypted data when the comparison result stored in the memory indicates a mismatch.

(89) Here, since the concepts of communication between the two semiconductor devices **1** and **2** according to the second embodiment are the same as those in FIG. **1** described above, illustration and explanation are omitted. FIG. **10** is a block diagram showing a configuration of a receiving-side semiconductor device **2***a* according to the second embodiment. The semiconductor device **2** a has substantially the same configuration as the semiconductor devices **1** and **2** described above, but the cryptographic unit **113***a* and the control unit **121***a* are changed.

(90) FIG. **11** is a block diagram showing an internal configuration of the cryptographic unit **113***a* according to the second embodiment. The cryptographic unit **113***a* includes an cryptographic processor **31***a*, an encryption processor **32***a*, and a comparison unit **33**. In the following description, the difference from the cryptographic unit **113** in FIG. **3** will be mainly described, and the description thereof will be omitted because the other configurations are the same.

(91) The cryptographic processor **31***a* is different from the cryptographic processor **31** in that it includes a data transfer unit **313***a*, a cryptographic circuit **314***a*, and a tag comparator **315**. In addition to the process of the data transfer unit **313**, the data transfer unit **313***a* acquires the authentication tag **452** from the memory **122** and outputs it to the tag comparator **315**. In addition to the process of the cryptographic circuit **314**, the cryptographic circuit **314***a* also outputs the generated verification tag to the tag comparator **315**. The tag comparator **315** compares the verification tag generated by the cryptographic circuit **314** with the authentication tag **452** acquired from the memory **122**, and determines whether or not they match. The tag comparator **315** outputs the determination result, that is, the tag comparison result, to the data transfer unit **313***a*, and stores the data in the memory **122** via the ECC/EDC **312**. In addition, the tag comparator **315** may determine that the tag comparison result is inconsistent with the tag comparison result, and output the result to the outside of the cryptographic unit **113***a*. The cryptographic circuit **314***a* may not

output the verification tag to the data transfer unit **313***a*. That is, the cryptographic unit **113***a* may not include the verification tag generated by the master in the processing result.
(92) Similarly, the cryptographic processor **32***a* includes a data transfer unit **323***a*, a cryptographic circuit **324***a*, and a tag comparator **325**, which are different from the cryptographic processor **32**. The data transfer unit **323***a*, the cryptographic circuit **324***a*, and the tag comparator **325** are substantially the same as the data transfer unit **313***a*, the cryptographic circuit **314***a*, and the tag comparator **315**. However, unlike the tag comparator **315**, the tag comparator **325** does not have to output the tag comparison result to the data transfer unit **313***a*.
(93) Returning to FIG. **10**, the description will be continued. The control unit **121***a* differs from the control unit **121** in the software to be executed, and does not compare the authentication tag with the verification tag. Instead, if the tag comparison result stored in the memory **122** indicates a mismatch, the control unit **121***a* discards the decoded data.
(94) The same configuration as that of the semiconductor device **2***a* can be used for the transmission-side semiconductor device according to the second embodiment.
(95) FIG. **12** is a diagram showing examples of data stored in the memory **122** in the receiving-side semiconductor device **2***a* according to the second embodiment. As a difference from FIG. **7**, the memory **122** includes a tag comparison result **463** in the processing result **46***a*. As described above, the tag comparison result **463** is a result of comparison between the authenticating tag **452** and the verifying tag by the tag comparator **315** in the cryptographic unit **113***a* of the semiconductor device **2***a*, for example, flag data.
(96) FIGS. **13** and **14** are sequence diagrams showing a flow of data reception, decoding, and verification processing according to the second embodiment. Here, since steps S**201** to S**208** in FIG. **8** are the same, illustration and explanation are omitted. After the step S**208**, the tag comparator **315** of the cryptographic processor **31***a* reads the authenticating tag **452** from the memory **122** according to the verifying tag outputted from the cryptographic processor **314***a* (S**310**). Then, the tag comparator **315** compares the authenticating tag **452** and the verifying tag, and determines whether or not they coincide with each other (S**311**). The tag comparator **315** outputs the tag comparison result to the data transfer unit **313***a*.
(97) Similarly, the tag comparator **325** of the cryptographic processor **32***a* reads the authenticating tag **452** from the memory **122** in response to the verifying tag outputted from the cryptographic processor **324***a* (S**312**). Then, the tag comparator **325** compares the authenticating tag **452** and the verifying tag, and determines whether or not they coincide with each other (S**313**). If, in a step S**311** or S**313**, a mismatch is determined, the tag comparator **315** or **325** may output to the outside of the cryptographic unit **113***a* that authentication of the received data has failed.
(98) Before, after, or in parallel with the S**313** from the step S**310**, the comparing unit **33** compares the processing results inputted from each of the cryptographic processors **31***a* and **32***a* (S**314**) in the same manner as in the step S**209**, and when the processing results do not coincide with each other, the comparing unit **33** outputs the fact that there is a failure to the outside of the cryptographic unit **113***a*.
(99) The cryptographic circuit **314** outputs the decryption data generated by the cryptographic circuit to the data transfer unit **313**. Therefore, the data transferring unit **313** stores the decrypted data acquired from the cryptographic circuit **314** and the tag comparing result acquired from the tag comparator **315** in the memory **122** via the ECC/EDC **312** (S**316**). After the cryptographic process by the cryptographic unit **113***a*, the secure CPU **111** reads the setting information from the cryptographic unit **113***a* and stores the setting information in the memory **122** (S**317**).
(100) Thereafter, the control unit **121** reads out the request information **47** and the setting information **48** from the memory **122** (S**318**), compares the request information **47** and the setting information **48** (S**319**), and determines whether or not they coincide with each other. When it is determined that they coincide with each other, the control unit **121** determines that the secure CPU **111** instructs the cryptographic process as requested by the request data **47**. If they do not coincide

in the S**319** of steps, a failure of the secure CPU **111** can be detected.
(101) Then, when it is determined in the S**319** of steps that the request information **47** matches the setting information **48**, the control unit **121** reads the tag comparison result **463** from the memory **122** (S**320**), and confirms the flag information indicated by the tag comparison result **463** (S**321**). That is, the control unit **121** checks whether or not the authentication tag **452** matches the verification tag generated by the cryptographic processing unit **31***a*. When the tag comparison result **463** indicates a match, the control unit **121** determines that the received data **45** has been authenticated. Therefore, the control unit **121** can proceed with the subsequent processing using the decoded data **461**. On the other hand, if they do not coincide with each other in the S**321** of steps, the received data **45** cannot be authenticated, so that the control unit **121** discards the decrypted data **461** and the like.
(102) As described above, in the second embodiment, the processing load of the software by the CPU can be reduced as compared with the first embodiment.
(103) Although the invention made by the inventor has been specifically described based on the embodiment, the present invention is not limited to the embodiment already described, and it is needless to say that various modifications can be made without departing from the gist thereof.

Claims

1. A semiconductor device comprising: a memory that stores data; a secure memory that stores a key used for encryption or decryption; a controller that 1) generates a cryptographic request for the data and 2) includes a master processor and a checker processor that execute a same process as one another in a lock step, a result of the checker processor being compared with a result of the master processor; a secure processor that sets setting information for the data based on the cryptographic request; a master cryptographic engine that includes a data transfer which transfers the data in the memory based on the setting information and a cryptographic circuit which performs a cryptographic process on the transferred data using the key; a checker cryptographic engine that includes a data transfer which transfers the data in the memory based on the setting information and a cryptographic circuit which performs a cryptographic process on the transferred data using the key based on the setting information; and a comparator that verifies the cryptographic processes performed by the master and checker cryptographic engines by comparing a result of the cryptographic process of the master cryptographic engine with a result of the cryptographic process of the checker cryptographic engine, wherein the setting information is checked by the controller after the cryptographic processes by the master and checker cryptographic engines are completed.
2. The semiconductor device according to claim 1, wherein the data transfer of the master cryptographic engine transfers the data without using the secure processor, and wherein the data transfer of the checker cryptographic engine transfers the data without using the secure processor.
3. The semiconductor device according to claim 1, wherein the data transfer of the master cryptographic engine transfers the data via a predetermined error detection mechanism, and wherein the data transfer of the checker cryptographic engine transfers the data via the predetermined error detection mechanism.
4. The semiconductor device according to claim 1, wherein the controller cannot access the secure memory.
5. The semiconductor device according to claim 1, wherein the setting information is checked by compared with a request information corresponding to the cryptographic request.
6. The semiconductor device according to claim 1, wherein the setting information includes an address of the data.
7. The semiconductor device according to claim 1, wherein the cryptographic circuit of the master

cryptographic engine performs an AES-GCM process, and wherein the cryptographic circuit of the checker cryptographic engine performs an AES-GCM process.